18. This follows at one from Exercise 17 and the fact that for abelian groups G_i, we have

$$(G_1 \times G_2 \times \cdots \times G_m)[p] = G_1[p] \times G_2[p] \times \cdots \times G_m[p].$$

This relation follows at once from the fact that computation in a direct prodct is performed in the component groups.

19. **a.** By Exercise 18, both m and n are $\log_p |T_p[p]|$.

b. Suppose that $r_1 < s_1$. Then the prime-power decomposition of the subgroup $p^{r_1}T_p$ computed using the first decomposition of T_p would have less than m factors, while the decomposition of the same subgroup computed using the second decomposition of T_p would still have $m = n$ factors. But applying Part(**a**) to this subgroup, we see that this is an impossible situation; the number of factors in the prime-power decomposition of an abelian p-power group H is well defined as $\log_p |H[p]|$. Thus $r_1 = s_1$.

Proceeding by induction, suppose that $r_i = s_i$ for all $i < j$, and suppose $r_j < s_j$. Multiplication of elements of T_p by p^{r_j} annihilates all components in the first decomposition given of T_p through at least component j, while the component $\mathbb{Z}_{p^{s_j}}$ of the second decomposition given is not annihilated. This would contradict the fact that by Part(**a**), the number of factors in the prime-power decomposition of any abelian p-power group, in particular of $p^{r_j}T_p$, is well defined. Thus $r_j = s_j$ and our induction proof is complete.

20. If $m = p_1^{r_1}p_2^{r_2}\cdots p_k^{r_k}$ for distinct primes p_i, then we know that $\mathbb{Z}_m \simeq \mathbb{Z}_{p_1{}^{r_1}} \times \mathbb{Z}_{p_2{}^{r_2}} \times \cdots \times \mathbb{Z}_{p_k{}^{r_k}}$ from Section 11. If we form this decomposition of each factor in a torsion-coefficient decomposition, we obtain the unique (up to order of factors) prime-power decomposition .

21. Following the notation defined in the exercise, we know that if the unique prime-power decomposition is formed from a torsion-coefficient decomposition, as described in Exercise 20, then cyclic factors of order $p_i^{h_i}$ must appear for each $i = 1, \cdots, t$. Because each torsion coefficient except the final one must divide the following one, the final one must contain as factors all these prime powers $p_i^{h_i}$ for $i = 1, \cdots, t$. Because the p_i for $i = 1, \cdots, t$ are the only primes that divide $|T|$, we see that m_r and n_r must both be equal to $p_1^{h_1}p_2^{h_2}\cdots p_t^{h_t}$.

22. If we cross off the last factors $\mathbb{Z}_{m_r}$ and $\mathbb{Z}_{n_r}$ in the two given torsion-coefficient decompositions of T, we get torsion-coefficient decompositions of isomorphic groups, because both decompositions must, by Exercises 19 and 20, be isomorphic to the group obtained by crossing off from the prime-power decomposition of T, one factor of order $p_i^{h_i}$ for $i = 1, \cdots, t$. (We are using the notation of Exercise 21 here.) We now apply the argument of Exercise 21 to these torsion-coefficient decompositions of this group, and deduce that $m_{r-1} = n_{r-1}$. Continuing to cross off identical final factors, we see that we must have the same number of factors, that is, $r = s$, and $m_{r-i} = n_{r-i}$ for $i = 0, \cdots, r-1$.

39. Free Groups

1. **a.** We obtain $a^2b^2a^3c^3b^{-2}$ whose inverse is $b^2c^{-3}a^{-3}b^{-2}a^{-2}$.

b. We obtain $a^{-1}b^3a^4c^6a^{-1}$ whose inverse is $ac^{-6}a^{-4}b^{-3}a$.

2. For the product in Part(**a**) of Exercise 1, it reduces to a^5c^3 and its inverse reduces to $a^{-5}c^{-3}$ in the abelian case. For Part(**b**), the product reduces to $a^2b^3c^6$ and its inverse reduces to $a^{-2}b^{-3}c^{-6}$.

3. a. There are $4 \cdot 4 = 16$ homomorphisms, because each of the two generator can be mapped into any one of four elements of $\mathbb{Z}_4$ by Theorem 39.12.

b. There are $6 \cdot 6 = 36$ homomorphisms by reasoning analagous to that in Part(**a**).

c. There are $6 \cdot 6 = 36$ homomorphisms by reasoning analagous to that in Part(**a**).

4. a. Let the free group have generators x and y. By Theorem 39.12, x and y can be mapped into any elements to give a homomorphism. The homomorphism will be onto $\mathbb{Z}_4$ if and only if not both x and y are mapped into the subgroup $\{0, 2\}$. Because 4 of the 16 possible homomorphisms map x and y into $\{0, 2\}$, there are 16 - 4 = 12 homomorphisms onto $\mathbb{Z}_4$.

b. Arguing as in Part(**a**), we eliminate the 4 homomorphisms that map x and y into $\{0, 3\}$ and the 9 that map x and y into $\{0, 2, 4\}$. The homomorphism mapping both x and y into $\{0\}$ is counted in both cases, so there are a total of 12 of the possible homomorphisms to eliminate, so 36 - 12 = 24 are onto $\mathbb{Z}_6$.

c. Arguing as in Part(**a**), we eliminate the 4 homomorphisms that map x and y into $\{\rho_0, \mu_1\}$, the 4 that map x and y into $\{\rho_0, \mu_2\}$, the 4 that map x and y into $\{\rho_0, \mu_3\}$, and the 9 that map x and y into $\{\rho_0, \rho_1, \rho_2\}$. The homomorphism that maps x and y into $\{0\}$ is counted four times, so we have found a total of 4 + 4 + 4 + 9 - 3 = 18 homomorphisms to eliminate, leaving 36 - 18 = 18 that are onto S_3.

5. a. There are 16 homomorphisms by the count in Exercise 3a.

b. There are 36 homomorphisms by count in Exercise 3b.

c. Because a homomorphic image of an abelian group is abelian, the image must be $\{0\}, \{\rho_0, \mu_1\}$, $\{\rho_0, \mu_2\}, \{\rho_0, \mu_3\}$, or $\{\rho_0, \rho_1, \rho_2\}$. The count made in Exercise 4c shows that there are 18 such homomorphisms.

6. a. There are 12 homomorphisms onto $\mathbb{Z}_4$ as in Exercise 4a.

b. There are 24 homomorphisms onto $\mathbb{Z}_6$ as in Exercise 3b.

c. There are no homomorphisms onto S_3, because the homomorphic image of an abelian group must be abelian, and S_3 is not abelian.

7. The definition is correct.

8. The definition is incorrect. The group must be free on the set of generators.

The **rank** of a free group G is the number of elements in a generating set A such that G is free on A.

9. Our reaction to these instances was given in the text. *You* have to give your reaction.

10. T F F T F F F T F T

11. a. We have $3(2) + 2(3) = 0$ but $3(2) \neq 0$ and $2(3) \neq 0$. A basis for $\mathbb{Z}_4$ is $\{1\}$.

b. We see that $\{1\}$ is a basis for $\mathbb{Z}_6$ because the group is cyclic with generator 1, and because $m1 = 0$ if and only if $m1 = 0$.

If $m_1 2 + m_2 3 = 0$ in $\mathbb{Z}_6$, then in $\mathbb{Z}$, we know that 6 divides $m_1 2 + m_2 3$. Thus 3 divides $m_1 2 + m_2 3$, and hence 3 divides $m_1 2$. Because 3 is prime and does not divide 2, it must be that 3 divides m_1. Thus 6 divides $m_1 2$ in $\mathbb{Z}$, so $m_1 2 = 0$ in $\mathbb{Z}_6$. A similar argument starting with the fact that 2 divides $m_1 2 + m_2 3$ shows that $m_2 3 = 0$ in $\mathbb{Z}_6$. Thus $\{2, 3\}$ is a basis for $\mathbb{Z}_6$.

c. Yes it is, for if x_i is an element of a basis of a free abelian group, then $n_i x_i = 0$ if and only if $n_i = 0$, so we stated the "independence condition" in that form there.

d. By Theorem 38.12, a finite abelian group G is isomorphic to a direct product $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_r}$ where m_i divides m_{i+1} for $i = 1, 2, \cdots, r-1$. Let b_i be the element of this direct product having 1 in the ith component and 0 in the other components. The computation by components in a direct product shows at once that $\{b_1, b_2, \cdots, b_r\}$ is a basis, and because the order of each b_i is m_i, we see that the orders have the desired divisibility property.

12. a. We proceed as suggested by the hint, and use the notation given there. Let $x \in G_1^*$. Because ϕ_1 is onto, we have $x = \phi_1(y)$ for some $y \in G$. Then $\theta_2(x) = \theta_2\phi_1(y) = \phi_2(y)$ and $\theta_1\theta_2(x) = \theta_1\phi_2(y) = \phi_1(y) = x$. In a similar fashion, starting with $z \in G_2^*$, we can show that $\theta_2\theta_1(z) = z$. Thus both $\theta_1\theta_2$ and $\theta_2\theta_1$ are identity maps. Because $\theta_2\theta_1$ is the identity, θ_2 must be an onto map and θ_1 must be one to one. Because $\theta_1\theta_2$ is the identity, θ_1 is an onto map and θ_2 is one to one. Thus both θ_1 and θ_2 are one to one and onto, and hence are isomorphisms. Thus G_1^* and G_2^* are isomorphic groups.

b. Let C be the commutator subgroup of G and let $\phi : G \to G/C$ be the canonical homomorphism (which we have usually called γ). Let $\psi : G \to G'$ be a homomorphism of G into an abelian group G'. Then the kernel K of ψ contains C by Theorem 15.20. Let $\theta : G/C \to G'$ be defined by $\theta(aC) = \psi(a)$ for $aC \in G/C$. Now θ is well defined, for if $bC = aC$, then $b = ac$ for some $c \in C$ and $\theta(bC) = \psi(b) = \psi(ac) = \psi(a)\psi(c) = \psi(a)e' = \psi(a)$ because C is contained in the kernel K of ψ. Now $\theta((aC)(bC)) = \theta((ab)C) = \psi(ab) = \psi(a)\psi(b) = \theta(aC)\theta(bC)$, so θ is a homomorphism. Finally, for $a \in G$, we have $\theta(\phi(a)) = \theta(aC) = \psi(a)$, so $\theta\phi = \psi$, which is the desired factorization. Thus $G^* = G/C$ is a blip group of G.

c. A blip group of G is isomorphic to the *abelianized version* of G, that is, to G modulo its commutator subgroup.

13. a. Consider the blop group G_1 on S and let G' be the free group $F[S]$, with $f : S \to F[S]$ given by $f(s) = s$ for $s \in S$. Because f is one to one and $f = \phi_f g_1$, we see that g_1 must be one to one. By a similar argument, g_2 must be one to one.

To see that $g_1[S]$ generates G_1, we take $G' = G_1$ and let $f(s) = g_1(s)$ for all $s \in S$. Clearly the identity map $\iota : G_1 \to G_1$ is a homomorphism and $f(s) = g_1(s) = \iota(g_1(s)) = (\iota g_1)(s)$, so $f = \iota g_1$, and the *unique* homomorphism ϕ_f mapping G_1 *into* G_1 is ι. Let H_1 be the subgroup of G_1 generated by $g_1[S]$. Thinking of H_1 as G' for a moment, we see that by hypothesis, there exists a homomorphism, $\phi_{H_1} : G_1 \to H_1$ and satisfying $f = \phi_{H_1} g_1$. Now ϕ_{H_1} also maps G_1 into G_1, and can also serve as the required homomorphism ϕ_f for the case where $G' = G_1$. By the uniqueness of ϕ_f, we see that $\phi_{H_1} = \iota$. Because $\phi_{H_1}[G_1] = H_1$ and $\iota[G_1] = G_1$, this can only be the case if $H_1 = G_1$. Therefore $g_1[S]$ generates G_1, and of course changing subscripts from 1 to 2 shows that $g_2[S]$ generates G_2.

Taking $G' = G_2$ and $f = g_2$, we obtain a homomorphism $\phi_{g_2} : G_1 \to G_2$ such that $\phi_{g_2} g_1 = g_2$. Taking $G' = G_1$ and $f = g_1$, we obtain a homomorphism $\phi_{g_1} : G_2 \to G_1$ such that $\phi_{g_1} g_2 = g_1$. Then for $s \in S$, we have $\phi_{g_1}\phi_{g_2} g_1(s) = \phi_{g_1} g_2(s) = g_1(s)$. Thus $\phi_{g_1}\phi_{g_2}$ is a homomorphism mapping G_1 into itself and acts as the identity on a generating set $g_1[S]$ of G_1, so it is the identity map of G_1 onto G_1. By a symmetric argument, $\phi_{g_2}\phi_{g_1}$ is the identity map of G_2 onto G_2. As in Exercise 12, we conclude that ϕ_{g_1} and ϕ_{g_2} are isomorphisms, so that $G_1 \simeq G_2$.

b. Let $G = F[S]$, the free group on S, and let $g : S \to G$ be defined by $g(s) = s$ for all $s \in S$. Let a group G' and a function $f : S \to G'$ be given. Let $\phi_f : G \to G'$ be the unique homomorphism given by Theorem 39.12 such that $\phi_f(s) = f(s)$. Then $\phi_f g(s) = \phi_f(s) = f(s)$ for all $s \in S$, so $\phi_f g = f$.

c. A blop group on S is isomorphic to the *free group* $F[S]$ on S.